Fundus Image Analysis for Age Related Retinal Disease Detection Using Deep Learning and XAI Methods

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Abstract- **Early recognition of age-related retinal conditions is crucial in avoiding irreversible loss of vision. This study presents a Deep Learning based diagnostic pipeline for classification of retinal fundus images. The RFMiD 1.0 and 2.0 datasets were merged and standardized, to produce a combined dataset consisting of 1098 fundus images covering ten different classes of age-related retinal diseases. All multilabel samples were discarded to keep a clean multiclass setting, and images were preprocessed via deterministic resizing, normalization, contrast enhancement and illumination correction. Trained four deep learning models ResNet-34, DenseNet-121, VGG-19 and Inception-V4. The models were evaluated using accuracy, precision, recall, F1-score and AUC score. Additionally, explainable AI techniques such as LIME, SHAP, Grad-CAM, and RISE were applied to interpret model predictions by highlighting pathology-relevant retinal regions. Among all the architectures, DenseNet-121 achieved the highest test accuracy of 88%, demonstrating the effectiveness of the proposed framework for reliable, interpretable, and scalable retinal disease screening.**

Keywords- **Retinal Fundus Imaging, Deep Learning, Age-related Eye Diseases, CNN, Explainable AI, Medical Image Classification.**

I. INTRODUCTION

Age-related retinal diseases including age-related macular degeneration (ARMD), diabetic retinopathy (DR), branch retinal vein occlusion (BRVO) and central Retinal Vein occlusion (CRVO) are major causes of vision loss throughout the world. These conditions progress gradually showing few or no symptoms in their early stages, making timely detection necessary for preventing long-term visual impairment. World Health Organization reports, more than 2.2 billion people globally experience near- or distance-vision impairment [1], and India alone contributes over 70 million people with such conditions [2]. With the rapid progress in deep learning and medical imaging, convolutional neural networks (CNNs) have been very successful in classifying complex retinal abnormalities from fundus images [3]. Retinal fundus imaging remains the most cost-effective approach to visualize essential retinal structures including macula, optic disc, and blood vessels for fast screening especially in low-resources situations [4].

In this study a single standardized experimental pipeline was designed to ensure a fair comparison between models. Four widely used CNN architectures—ResNet-34 [5], DenseNet-121 [6], VGG-19 [7], and Inception-V4 [8]—were trained on combined RFMiD dataset under identical experimental settings.

II. RELATED WORKS

There has been extensive research on deep learning-based methods of automatic detection of retinal fundus diseases. The work presented in [9] highlighted the fragmented nature of existing retinal disease research. Their survey summarized progress and comparison of popular architectures including VGG, ResNet, FCN, and U-Net, and reported results for well-studied diseases such as diabetic retinopathy (95% accuracy) and glaucoma detection (0.97 AUC) on datasets like MESSIDOR, REFUGE, AREDS, and ODIR. The authors also pointed out that there is a clear gap in research for rare retinal conditions, and there is an increasing need for explainable AI

systems and more balanced datasets. To address dataset inconsistencies, [10] proposed a self-supervised learning on the AREDS dataset for age-related macular degeneration (AMD). Their method reduced the dependency on manual annotations and achieved a balanced accuracy of 82% and screening accuracy of 87% for referable AMD.

To deal with the computational cost of Deep Learning Models, [11] presented a lightweight CNN with ten convolutional layers. The model achieved 95% accuracy on the EyeNet dataset. This demonstrates the feasibility of retinal diagnostics on resource-limited hardware. Another study [12] proposed a hybrid Generative Adversarial Network (GAN)/CNN transfer learning pipeline to predict AMD progression and achieved 93% sensitivity and 88% specificity values. The results showed the use of multi-stage deep learning models in clinical decision support. In [13], transfer learning and federated learning were combined for dry eye disease detection, ensuring data privacy while achieving an accuracy of 84.62%. A review presented in [14] discussed a range of AI-driven medical imaging systems for detection of DR and AMD, reported that the ResNet and U-Net variants exceed 90% accuracy. Another study [15] examined drusen (DN) pattern detection using CNNs, Recurrent Neural Networks (RNNs), and U-Net, reporting almost perfect results on the AREDS and proposed the importance of larger annotated datasets for real-world deployment. [16] reported AUC scores close to 99% for AMD and DR using Vision Transformers and ensemble-based classification methods.

Altogether, these works demonstrates that the Deep Learning models are effective in the classification of retinal fundus images. Despite the tremendous successes, the issue of imbalanced datasets and model interpretability still exists. This limitation motivates the present study, which focuses on data augmentation and consistent evaluation across multiple CNN architectures to improve fairness and robustness in multi-class retinal disease classification.

III. METHODOLOGY

The process pipeline for classification of retinal fundus images is shown in Fig.1. The components of the pipeline are discussed in the following subsections.

A. Dataset Merging

Two publically available retinal fundus image databases RFMiD1.0 [17] and RFMiD2.0 [18] were combined, which contains high resolution images captured under clinical settings and include expert-verified diagnostic label. To maintain consistency across the dataset, all metadata fields and disease annotations were standardized and naming conventions were unified.

B. Dataset Cleaning and Split

Since the objective of this study was to perform single-label multi-class classification, images containing multiple co-existing retinal conditions were dropped. After this dataset cleaning, the final dataset consisted of 1,098 images belonging to ten age-related retinal disease classes. The ten classes were labelled as: Anterior Ischemic Optic Neuropathy (AION), ARMD, BRVO, CRVO, DN, DR, Epiretinal Membrane (ERM), MHL (Macular Hole), Optic Disc Cupping (ODC) and Retinal Pigment Epithelium Changes (RPEC). The dataset was divided in 70:15:15 ratio, resulting in 768 training images, 165 validation images, and 165 test images. Table I shows number of images in train, validation and test set.

C. Preprocessing Pipeline

All the images were resized to a uniform size to fit the model input. Normalization and illumination was done to scale pixel values using ImageNet statistics and reduce lighting variation in retinal images respectively.

D. Training Pipeline

Baseline training (without augmentation): The model was trained on raw dataset without doing any image transformations The main purpose for doing this was to establish a baseline performance for comparison.

Hybrid augmentation (Fixed+Online): Data augmentation increased dataset diversity by transforming the images. Fixed augmentation strategy was combined with online augmentation so that the minority disease classes are not unrepresented.

Table I**:** Per class distribution of fundus images

Class	Train Count	Val Count	Test Count	Augmented Train Count
AION	15	4	3	45
ARMD	52	12	11	104
BRVO	69	15	15	69
CRVO	35	7	8	70
DN	111	23	24	111
DR	329	71	71	329
ERM	11	2	2	33
MHL	8	2	2	24
ODC	120	25	26	120
RPEC	18	4	3	54

E. Deep Learning Models Training

Four deep learning networks (ResNet34, VGG19, DenseNet121 and Inception-v4) were finetuned for the classification of retinal fundus images in identical training settings. All the models were first trained for 50 epochs with a fixed learning rate of 1×10^{-4}, batch size of 32 using AdamW optimizer and CrossEntropyLoss. All of the model's experiments were conducted with identical pre-processing and data set splits. These architectures are mainly different in terms of depth, connectivity pattern, input size constraints and the cost of computation.

Table II: Model parameters before augmentation

Model	Input Size	Architecture Type	Parameters
ResNet34	224 × 224	Residual (skip connections)	~21.8M
VGG19	224 × 224	Deep sequential conv layers	143.7M
DenseNet121	224 × 224	Dense connections (feature reuse)	~7.9M
Inception-v4	299 × 299	Multi-branch Inception modules	~42M

To improve the class balance in the training set, particularly for minority disease classes such as AION, ERM, MHL, and RPEC fixed rotation-based augmentation (±9°) was applied. This procedure expanded AION, ERM, MHL, and RPEC by 200%, while moderately represented categories such as ARMD and CRVO were increased by 100%. The remaining well-represented classes such as BRVO, DN, DR, and ODC—were left unaugmented to avoid unnecessary redundancy. After fixed augmentation, the training set increased from 768 to 959 images. Table I shows number of images in final augmented training set.

Online augmentations includes random horizontal flipping, ±20° rotations, and color jittering—were applied dynamically to introduce continuous variation in image appearance and prevent overfitting. During training, online augmentation was applied for training and validation images. This hybrid (Fixed+Online) augmentation strategy ensured that the models were exposed to balanced class distribution and diverse transformations on images.Model parameters used to train the model before augmentation is given in Table II and after augmentation is given in Table III.

Table III: Model parameters after augmentation

Parameter	ResNet34	VGG19	DenseNet121	Inception-v4
Input Size	384 × 384	224 × 224	384 × 384	299 × 299
Architecture Type	Residual	Deep Fully Connected (FC) Layer + Conv Stack	Dense Connections	Multi-branch Inception
Training Strategy	Full fine-tuning	2-Stage (head → full)	Partial fine-tuning (last block)	Full fine-tuning
Regularization Used	Dropout(0.4)	VGG classifier + Dropout	BN + Dropout(0.5)	Dropout(0.4)
Classifier Head	Dropout → Linear	Large FC Stack	Batch Normalization (BN)→ Dropout → Linear	Dropout → Linear
Parameters (Approx.)	~21M	143M	~8M	~42M

F. Evaluation Matrix

Certain metrics were used to check how well the model performs: Accuracy that gives percentage of correctly classified images, Precision that tells how reliable positive prediction are there, Recall that tells how many disease classes are correctly identified and F1-score that gives the balanced measure of precision and recall. Macro-Average F1 score to evaluate models performance across all 10 classes. AUC-ROC (Area Under Curve-Receiver Operating Characteristic) score to evaluate the sensitivity of the classification model.

G. Explainability

XAI methods majorly identifies the regions in fundus images that influenced the prediction. For each of the four models, four different XAI techniques were used.

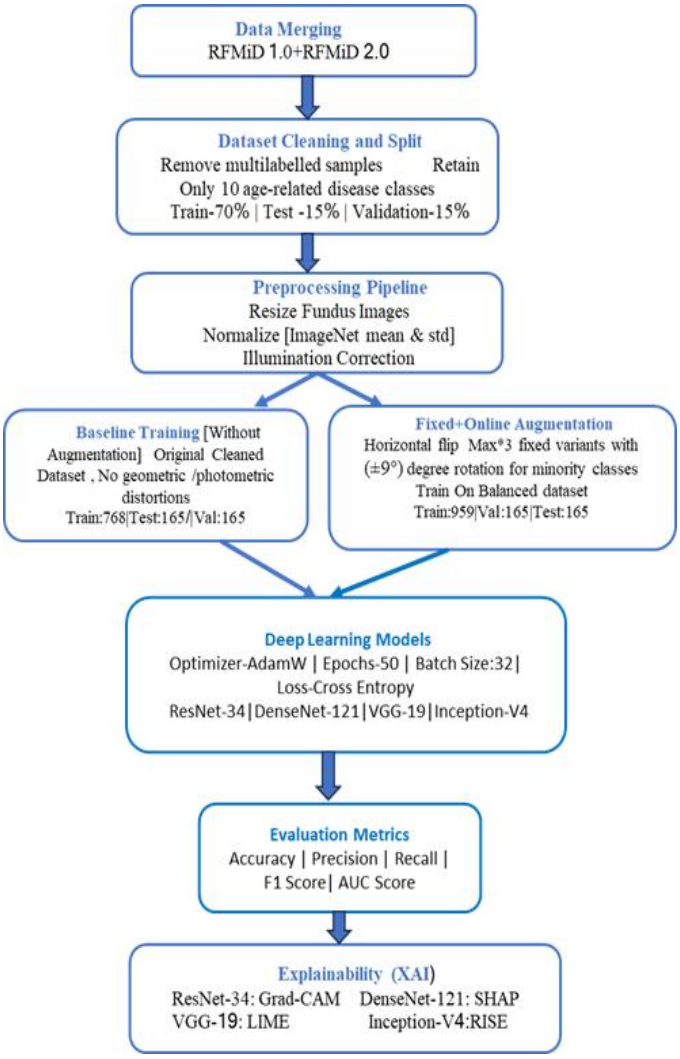


Fig 1: Process pipeline for classification of retinal fundus images

IV. RESULTS AND DISCUSSION

This section presents in-depth analysis of four deep learning architectures—ResNet34, DenseNet121, Inception-v4, and VGG19—before and after applying the augmentation. The analysis highlights the influence of dataset imbalance, the effect of augmentation, and the final comparative performance of all the models.

Table IV: Overall performance comparison of four models without augmentation

Model	Accuracy (%)	Weighted Precision (%)	Weighted Recall (%)	Weighted F1-Score (%)	Macro F1-Score
ResNet-34	70.91	66.68	70.91	67.97	0.3848
DenseNet-121	72.73	71.75	72.73	71.58	0.5123
VGG-19	77.58	75.48	77.58	76.14	0.4608
Inception-v4	**81.82**	**80.81**	**81.82**	**80.96**	**0.5657**

Table IV summarizes the baseline results obtained from original imbalanced dataset before applying augmentation. Inception-v4 performed the best among all the models with test accuracy of 81.82%. It showed strong performance on high-frequency classes including DR, ODC and ARMD with F1-scores above 0.80. Due to imbalance in the RFMiD dataset, all models failed on rare disease classes like ERM and RPEC. Validation curves confirmed early and strong overfitting particularly in VGG19 and ResNet34, where the validation accuracy ceases to increase and fluctuates across the epochs. Overall, the pre-augmentation performance revealed the inability of all models to generalize well across minority disease categories and models were biased towards majority classes.

Table V: Overall performance of four models with augmentation

Model	Accuracy (%)	Weighted Precision (%)	Weighted Recall (%)	Weighted F1 (%)	Macro F1
ResNet-34	**85.45**	83.52	85.45	84.14	0.6480
DenseNet-121	**88.48**	**88.74**	**88.48**	**87.32**	**0.6974**
VGG-19	**76.97**	79.83	76.97	77.34	0.5991
Inception-v4	**83.03**	82.26	83.03	81.97	0.5957

After applying the fixed rotation-based (±9°) augmentation combined with online augmentation (flips, jitter, ±20° rotation), all models achieved substantial improvements in class-wise fairness, stability, and generalization. As shown in the Table V, ResNet34 showed improvement, with test accuracy rising from 70.91% → 85.45% and macro-F1 increasing from 0.38 → 0.6480. DenseNet121 achieved the best overall performance, improving test accuracy from 72.73% → 88.48% with macro-F1 rising from 0.5123 → 0.6974, and as shown in the Fig. 2 and 3 previously undetectable classes like ERM and RPEC became learnable. Inception-v4, which already performed well before augmentation, showed a modest accuracy gain of +1.21% and exhibited marked improvements in stability. VGG19 benefited substantially from reduced overfitting and smoother convergence, reaching 77.00% accuracy and increasing macro-F1 from 0.4608 → 0.599.

The results confirm that, addressing class imbalance is essential for developing clinically reliable deep learning systems. Fig. 4 shows that AUC-ROC curve for DenseNet-121 model which achieved a micro-Average AUC score of 97.7%. The training and validation loss curve is given in Fig. 5 for DenseNet-121. Similarly, training and validation accuracy curve is given in Fig. 6. The analysis of the curves reveals that the model is able to learn ten disease classes data appropriately. F1 score for each class before and after augmentation is shown in Table VI. Classes ERM and RPEC achieved increased F1 score of 0.6667 and 0.500 after augmentation. But for MHL class there was a decrease in F1 score after augmentation.

Table VI: Per class F1-Score of DenseNet-121

Classes	F1-Score (Before Augmentation)	F1-Score (After Augmentation)
AION	0.5000	0.5000
ARMD	0.4762	0.8421
BRVO	0.5926	0.8462
CRVO	0.8750	0.9412
DR	0.8652	0.9396
MHL	0.5000	0.0000
ERM	0.0000	0.6667
ODC	0.7541	0.9231
RPEC	0.0000	0.5000
DN	0.5600	0.8148

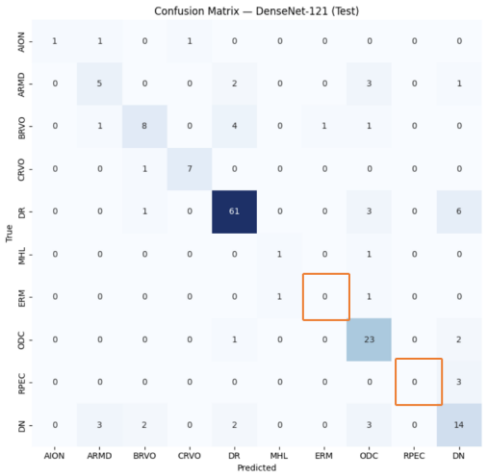


Figure 2: Confusion Matrix for Dense-121 Before Augmentation

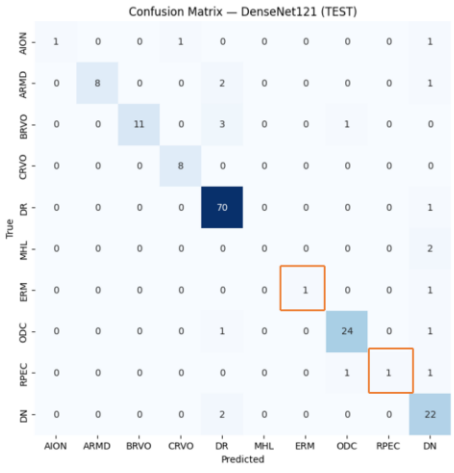


Figure 3: Confusion Matrix for DenseNet-121 After Augmentation

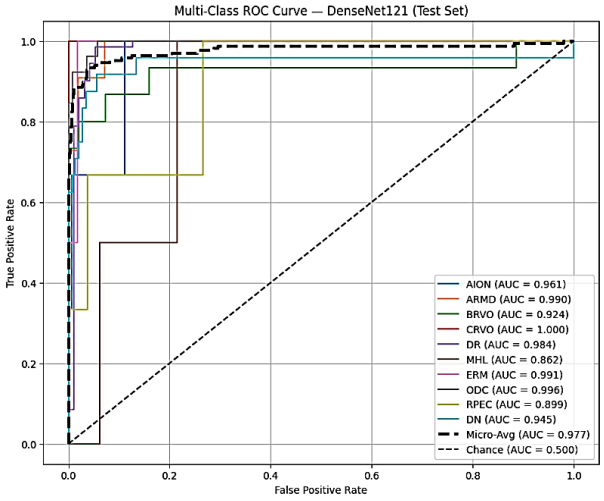


Figure 4: ROC-AUC Curve for DenseNet-121

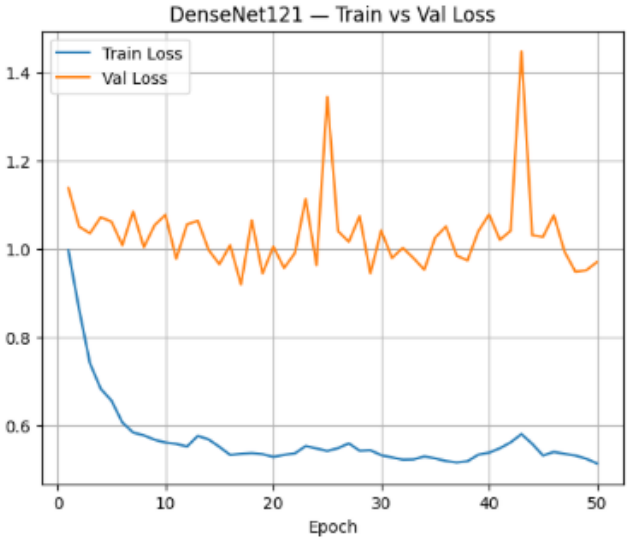


Figure 5: Training And Validation Loss Curve for DenseNet121

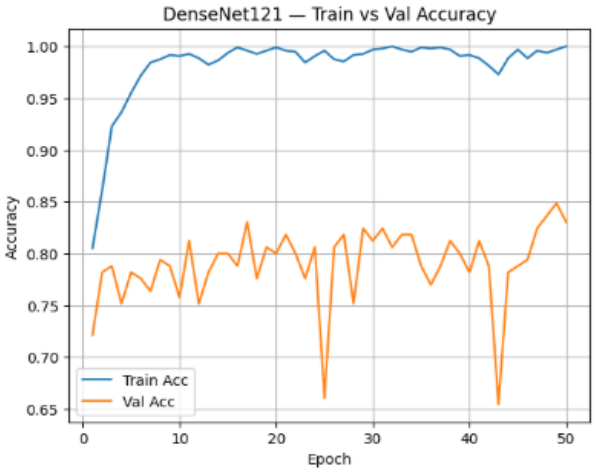


Figure 6: Training And Validation Accuracy Curve for DenseNet121

A. Explainable AI (XAI) Integration

For improving interpretability XAI methods were used, including LIME for VGG-19, SHAP for DenseNet-121, RISE for Inception-V4, and Grad-CAM for ResNet-34. The visual explanation of all four techniques highlighted the retinal areas that majorly contributed toward the predictions of each model. Fig. 7 shows XAI methods interpretation for four models for a sample image of DR class.

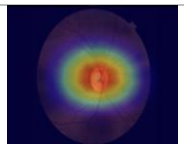
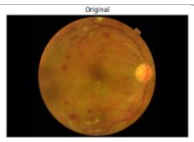
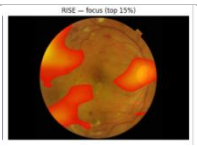
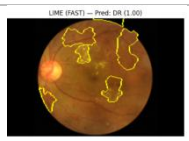
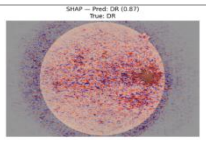
Model	XAI	Original Image	XAI Image
Resnet34	Grad-CAM		
InceptionV4	RISE XAI		
VGG19	LIME		
Densenet121	SHAP XAI		

Figure 7: Sample XAI images for four models

V. CONCLUSION AND FUTURE WORK

In this work, we explored Deep Learning application for detection of ten age-related retinal diseases using fundus images. By applying hybrid augmentation (Fixed+Online). Four CNN models ResNet-34, DenseNet-121, VGG-19, and Inception-V4 were trained under identical experimental settings.

The results highlight the importance of managing dataset imbalance. Without augmentation, models could not predict rare diseases. After applying the hybrid augmentation strategy, every model showed improvements in generalization and minority-class detection. DenseNet-121 achieved the best results with 88% test accuracy and ResNet-34 85.45%, Inception-V4 83.03% and VGG-19 benefited from reduced overfitting and achieved 77% test accuracy.

To demonstrate the areas of the retina that the model focussed on, XAI techniques like LIME, SHAP, RISE, and Grad-CAM were implemented. XAI's can bridge the gap existing between algorithmic decisions and interpretability, so that the system becomes more transparent and acceptable for medical practice.

The classification result and explainability of all four models can be provided to a Large Language Models (LLM) to generate diagnosis report, which can be looked forward as a future work.

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